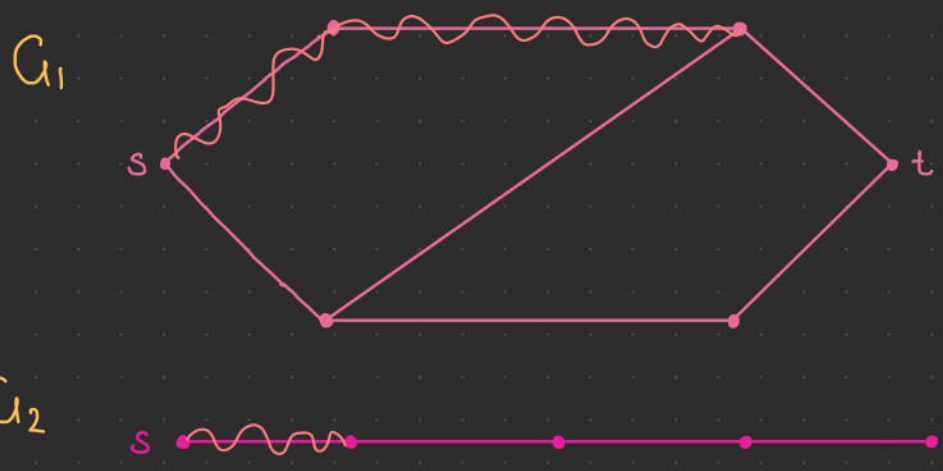
Topic 2: Connectivity



G_1 is "more connected" than G_2

Let $G = (V, E)$ be a connected graph. A <u>vertex cut</u> of G, is a set of vertices $S \subset V$ s.t $G - S$ has more than one component (disconnected).

The (vertex) <u>connectivity</u> of a graph G, $\kappa(G)$, is the minimum size of a vertex set S s.t $G-S$ disconnects G, or is a single vertex. {this becomes important for connected graphs, which will hang on till the end.}

If $\kappa(G) \geq k$, then G is <u>k-connected</u>.
(cappa; letter k)

{This means that k-connected graphs are also ℓ-connected for $1 \leq \ell \leq k$}

If G is disconnected, then $\kappa(G) = 0$.

G has connectivity 1 iff it is a graph with a cut vertex (contains no cycles, is a tree).

e.g. 2.1

(i) The complete graph K_n is $(n-1)$ connected


K_4


K_5

$\kappa(K_n) = n-1$

$(n-2)$-connected too

⋮

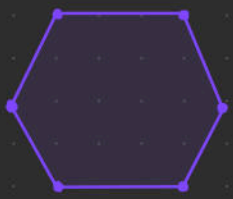
all the way to <u>1</u>-connected

(ii) Trees are 1-connected



$\kappa(T) = 1$

(iii) Cycles are 2-connected ($|V| \geq 3$)



1 - connected

2 - connected

(iv) The bipartite graph, $K_{m,n}$, is m-connected ($1 \leq m \leq n$)



1 - connected

$\kappa(K_{2,3}) = 2$

In general, if $|V| = n$, we have:

(disconnected) $0 \leq \kappa(G) \leq n-1$ (complete),

and for a vertex v,

$\kappa(G) - 1 \leq \kappa(G - v)$ { it depends where you cut from

In other words, deleting a vertex can only ever decrease the connectivity of a graph by at most 1.

think: how many vertices do I need to take out to disconnect the graph?

Measure of connectivity:

The minimum # of vertices that must be deleted from G to disconnect it (or produce K_1).

Menger's Theorem {Cornerstone of graph theory}

In this section, we look at another way to measure connectivity in a very important theorem.

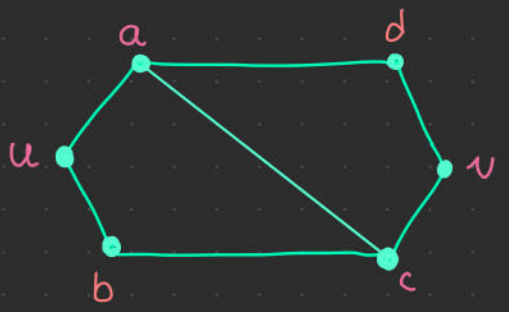
Given graph $G = (V, E)$, a set $S \subset V$ is said to separate two vertices u & v of G if

* $G - S$ is disconnected, and
* u & v belong to different components of $G - S$.

S is called a uv-separating set

* mixing ways to communicate, sending a mobile phone on a carrier pidgeon.
* when there are multiple ways for the nodes to connect, and they start to mix.

Example 2.2



① Find all uv-paths.
② Find all uv-separating sets
③ How are they related?

$S = \{a, b\}$ is a cut set of G.
S is a u-v separating set.

Observations

If S seperates u and v, then

i) S is a vertex cut of G → This means that $\kappa(G) \leq |S|$
ii) u and v cannot be adjacent



Restated Menger's Theorem:

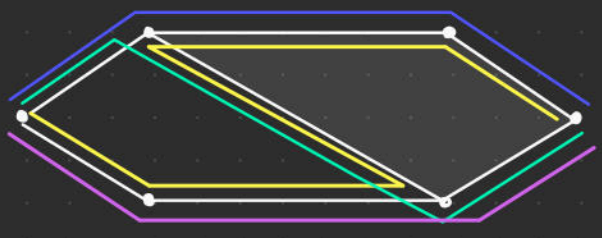
For any two distinct vertices u and v in a graph G, the smallest number of vertices whose removal separates u from v is exactly equal to the maximum number of internally vertex-disjoint paths s.t

> $\kappa(G) \geq k$ iff for every distinct pair of vertices, $\exists$ at least k internally vertex-disjoint paths connecting them.

Note: By internally vertex-disjoint paths, we mean a set of paths in a graph that share no common vertices other than their initial and final verts.

> Suppose distinct vertices u and v in a graph G. Then a collection of paths from u to v is internally disjoint if each vertex of the graph (apart from u and v) appears on at most one of these paths.

Example 2.3 — recall from example 2.2



P_1
P_2
P_3
P_4

P_1 and P_4 are internally disjoint paths.

A collection $\{P_1, P_2, \ldots, P_k\}$ of u-v paths in a group G is internally disjoint in which every pair of paths have only vertices u and v in common.

Theorem 2.1 (Menger's Theorem)

Let u and v be non-adjacent vertices. The minimum number of vertices in a u-v separating set equals the maximum number of u-v internally disjoint paths.

Exercise 2.4: In ex 2.2 and 2.3, we saw that $|s|$ is at least 2 and there are at most 2 internally disjoint uv paths.

This then gives a simple characterisation of separating sets.

Theorem 2.2

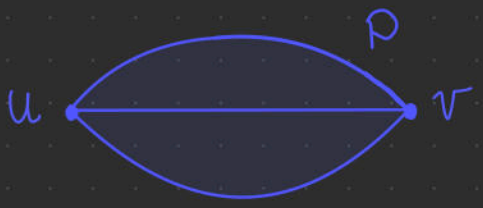
Let u and v be two distinct vertices of a graph $G = (V, E)$. A set $S \subseteq V$ separates u and v with $u, v \notin S$ iff every u-v path has at least one internal vertex belonging to S.

$$A \longleftrightarrow B \equiv (A \to B) \wedge (B \to A)$$

proof: $(A \to B) \equiv (A \wedge \neg B) \to \mathbb{F}$

Assume S is a u-v separating set. Let P be a u-v path in G.

Quick note on proof construction:
If you say let P be something and want to add another property to P, dont say let again, say suppose.

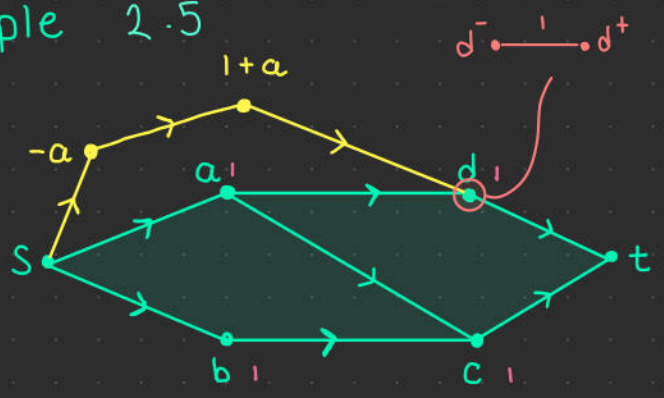


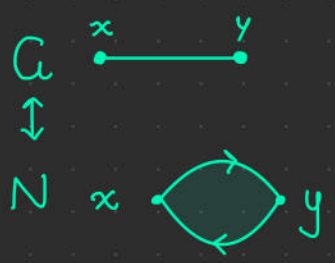
Suppose P has no internal vertices belonging to S. Then deleting S leaves P in tact. So u and v will be in the same connected component of $G-S$, which is a contradiction (of our assumption), as S is a u-v separating set. Hence every u-v separating path must have at least one vertex belonging to S.

converse: ($\leftarrow$) B $\rightarrow$ A

Suppose that every u-v path has an internal vertex belonging to S. Then, if we delete S from G, there will be no paths between u and v. That is, they are in different components of $G-S$, and hence S separates u and v. □

Example 2.5





★ How do I convert G to N?

★ How do I assign capacities to edges in N to give s.t vertex disjoint paths in G?

★ Use theorems (somehow) to 'pick out' vertex disjoint path

★ But we have capacity on vertices, networks have

capacities on edges.



Vertex Capacity ⟶ Edge capacity (network)

G ⇄ N

Proof: main idea: $(a \geq b) \wedge (b \geq a) \Rightarrow a = b$.

Let $P_1, \ldots, P_m$ be a set of m internally disjoint u-v paths and assume that m is the maximum size of such a set. Let S be a u-v separating set of minimum size and let $|S| = n$. By thm. 2.2, each of the paths $P_1, \ldots, P_m$ contains at least one member of S. Since the paths are internally disjoint, no member of S can occur in more than one path. It follows that we have at least as many vertices in S as we have internally disjoint u-v paths. Hence $m \leq n$ e.g. ①

We are going to convert a flow network N

to do this:

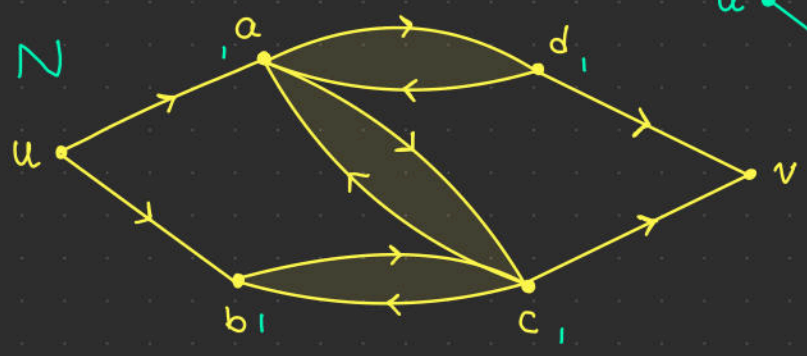
~ ★ Make u the source and v the sink

★ Direct all incident edges away from u.

★ Direct all incident edges towards v

~ ★ Replace each edge xy $(x, y \in V \mid \{u, v\})$ by two directed edges xy and yx.

Working example:







$G \to N \to N'$

★ Assign each vertex (other than u and v), capacity 1.

We can then convert $N \to N'$, an equivalent network in which only edges have capacities.

Then we will consider a maximum flow on N' and show that $m = val(f)$

We start by replacing every vertex $x \in N$ with vertices x^- and x^+ (except for u and v).

Then join x^- to x^+ within the edge xx^+



The edges are called internal edges and all other edges external. Internal edges are assigned capacity 1 and all external edges have capacity 0.

Every vertex of N' (except u an v) is of the form x^- and x^+ : the start and end of exactly one internal edge.

It follows that the value of any flow into or out of any vertex in N' is either 1 or 0. Hence f

assigns a value of either 0 or 1 to each edge of N'. This means that f must use $val(f)$ of internally disjoint paths in N', each contributing 1 to the value of f. Since each of these internally disjoint $u-v$ paths in N' corresp. to an internally disjoint $u-v$ path in G, then $val(f) \leq m$.

There are m internally disjoint $u-v$ paths which can carry a flow of 1 in N', hence $m \leq val(f)$.

It follows that $m = val(f)$.

Hence, can we find a min cut?
How do you form a cut set?

When we find a max flow, we end up with a tree full of partial maximal* paths



WTS: $m = n$ (menger's thm) $\iff (n \geq m) \wedge (m \geq n)$

n = size of min $u-v$ cut set

m = # of internal distinct $u-v$ paths

$m \leq n$ from eq ①, $m = val(f) \geq n$ from eq ②

By Thm 1.4 (MFMC thm.), there exists a minimum cut $(U, \bar{U})$ in N' s.t $c(U, \bar{U}) = val(f)$.

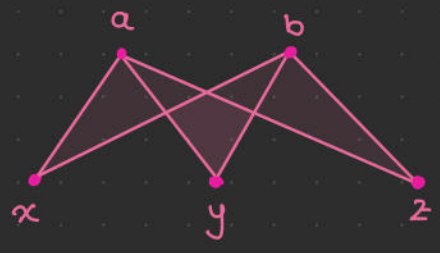
We will show that the capacity of this cut is $\geq n$, s.t $c(U, \bar{U}) \geq n$ {the size of the minimuum u-v separating set of G}. Since $c(U, \bar{U}) = val(f)$, and $val(f)$ is finite, then each edge from U to $\bar{U}$ must be an internal edge.

Moreover, since $(U, \bar{U})$ is a cut, each directed path from u to v must use at least one of these edges. Hence, deleting either all of the 'heads' or 'tails' of these edges removes all possible u-v paths from N'. Hence, the set of vertices in G corresponding to these heads or tails (depending on which one was chosen) corresponds to a u-v separating set of size $val(f)$. Therefore $val(f) \geq n$.

It follows from ②, that $m = val(f) \geq n$. Recall from ①, that $m \leq n$. Hence $m = n$, as required.

Example 2.7

Consider $K_{2,3}$



Remember that Menger's theorem is only for non-adjacent vertices only.

Consider a and b. There are 3 internally disjoint a-b paths. The set $\{x, y, z\}$ is an a-b sep. set of size 3. (ie. $|S| = 3$).

Consider x and y.

☆ There are 2 internally disjoint x-y paths

★ The set $\{a, b\}$ is an x-y sep. set of size 2.

Note: Menger's thm useful for connectivity between vertices, less so for whole graph.

Theorem 2.3 (Whitney characterisation k-connected)
A graph G is k-connected if and only if for each pair of distinct vertices $u, v \in G$, there are at least k internally dist. u-v paths in G.

Recall that we defined $\kappa(G)$, the connectivity of a graph G as

* $\kappa(G)$ is the min # vertices that must be deleted from G to disconnect it (or produce K_1).
* $\kappa(G)$ is the max k for which every pair of vertices is connected by k internally disjoint paths

proof: ($\rightarrow$)
Suppose that G is a k-connected graph, and u, v are distinct vertices of G.

There are two cases to consider.
Case 1: Suppose that u and v are not adjacent, and let S be a u-v separating set. Since G is k-connected,

$$|S| \geq |\kappa(G)| \geq k.$$

Then by Menger's Thm., G contains at least k internally disjoint u-v paths.

Case 2: Suppose that u and v are adjacent, and let $e = uv$. Then $G - e$ is $(k-1)$-connected {may also be k-connected, but true to say $k-1$ con.}. Let S_2 be a minimum u-v separating set in $G-e$. Thus

$$|S_2| \geq |\kappa(G-e)| \geq k-1.$$

By Menger's thm (since we deleted e, we can use this), $G-e$ contains at least $k-1$ internally dist. u-v paths. Hence G contains at least k internally disjoint paths (since e is one such path). ($\leftarrow$)

Edge Connectivity

A cut edge of our graph, $G = (V, E)$, is an edge cut of the form $S = (U, \bar{U})$ where $U \subset V$ and $U \neq \emptyset$ s.t $G-S$ has more than one component

The edge connectivity of G, $\lambda(G)$, is the minimum size of an edge cut.

If $\lambda(G) \geq k$, then G is k-edge-connected
i.e. there does not exist $k-1$ edges that disconnects G.

Note that $\lambda(G) = 0$ iff G is disconnected. Similarly:

$\kappa(K_n) = (n-1) = \lambda(K_n)$

$\kappa(T) = 1 = \lambda(T)$

$\kappa(C_n) = 2 = \lambda(C_n)$

$\kappa(K_{m,n}) = \min(m,n) = \lambda(K_{m,n})$

★ Not the same as vertex connectivity

G  In G, $\kappa(G) = 1$, $\lambda(G) = 2$.

★ So what is the relationship between them?

Theorem 2.4: For every graph G with min degree $\delta(G)$,

$$\kappa(G) \leq \lambda(G) \leq \delta(G).$$

(proof)

The edges incident to a vertex v in G of min. degree form an edge cut. Hence $\lambda(G) \leq \delta(G)$.

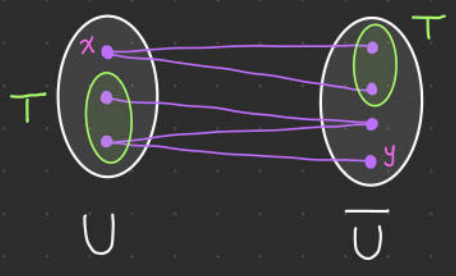
It remains to show that $\kappa(G) \leq \lambda(G)$. Let $(U, \bar{U})$ be a minimum edge cut of G, and recall that $\kappa(G) \leq (n-1)$. If every vertex in U is adjacent to every vertex in $\bar{U}$, then

$$|(U, \bar{U})| = |U| \cdot |\bar{U}| \geq (n-1) \geq \kappa(G).$$

Hence $\lambda(G) \geq \kappa(G)$.

Otherwise, we choose $x \in U$ and $y \in \bar{U}$ with x not adjacent to y (such a pair will always exist).

Then $|(U, \bar{U})| = \lambda(G)$ (by def).



Let T be the set of all vertices adjacent to $x \in U$ and all vertices in $U - \{x\}$ adjacent to vertices in $\bar{U}$.

Every xy-path passes through T, so T is an xy-

separating set, and $|T| \geq \kappa(G)$ by def.

If we choose the edges from x to $T \cap \bar{U}$ and one edge from each vertex of $T \cap U$ to $\bar{U}$, we have $|T|$ edges of this cut. Hence $|T| \leq |(U, \bar{U})|$.

Thus $\kappa(G) \leq |T| \leq |(U, \bar{U})| = \lambda(G)$ by def., and so we have that $\kappa(G) \leq \lambda(G)$.

Connecting this all to Menger's theorem:

Let u, v be vertices in a graph $G = (V, E)$. A collection of u-v paths $\{P_1, P_2, \ldots, P_m\}$ is edge-disjoint if no pair of paths have an edge in common.



Theorem 2.5 (Menger's Theorem for edge connectivity)

Let u and v be distinct vertices in a graph (note we dont require non adjacency). Then the minimum number of edges separating u from v equals the maximum number of edge-disjoint u-v paths.

tool for :☆: edge connectivity with pairs of vertices

not the tool for more general than that.

Q : Can we 'recycle' the edge version of the proof?

Thm 2.6 (Whitney)

A (simple) graph G is k-edge-connected if and only if for each pair of distinct vertices u and v in G, there are at least k edge-disjoint u-v paths in G.

{constructive proof}{splitting vertices into sets}

(Rewritten proof of thm 2.4 ↓ ↓ ↓)

proof: (thm 2.4)

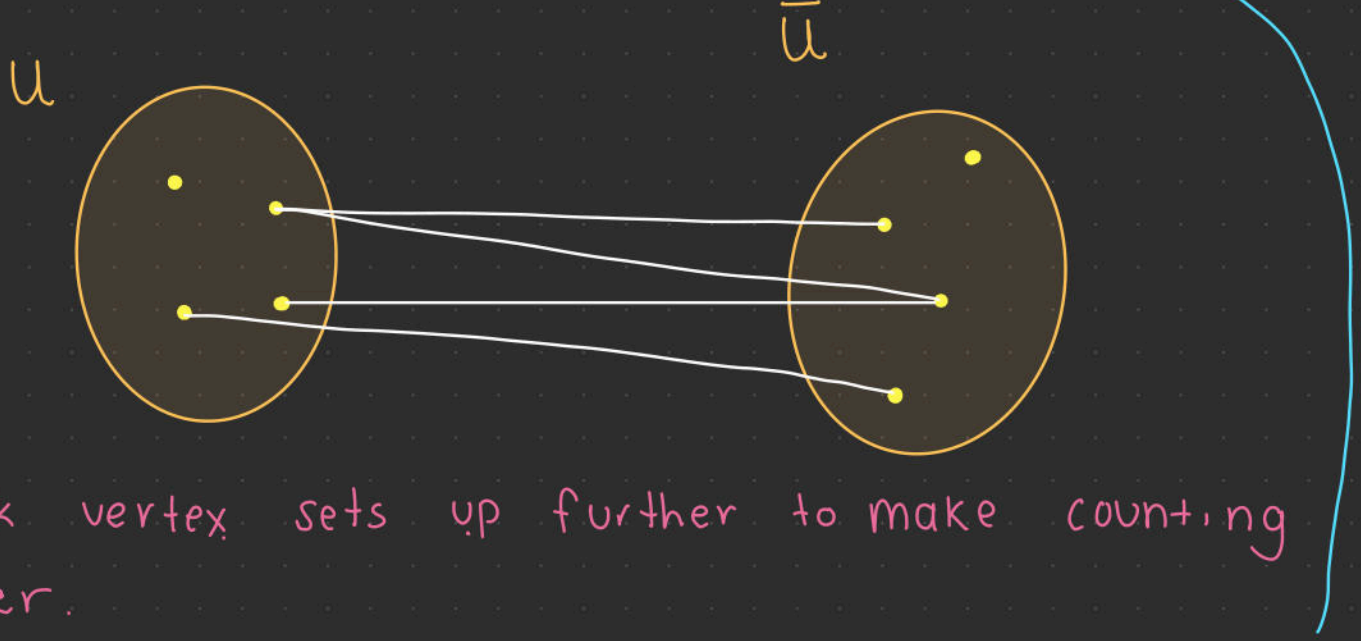
The edges incident to a vertex v of min. deg. form an edge cut, hence



$$\lambda(G) \leq \delta(G)$$

It remains to show that $\kappa(G) \leq \lambda(G)$.

Let $(U, \bar{U})$ be a minimum edge cut of G, and recall that $\kappa(G) \leq n-1$.



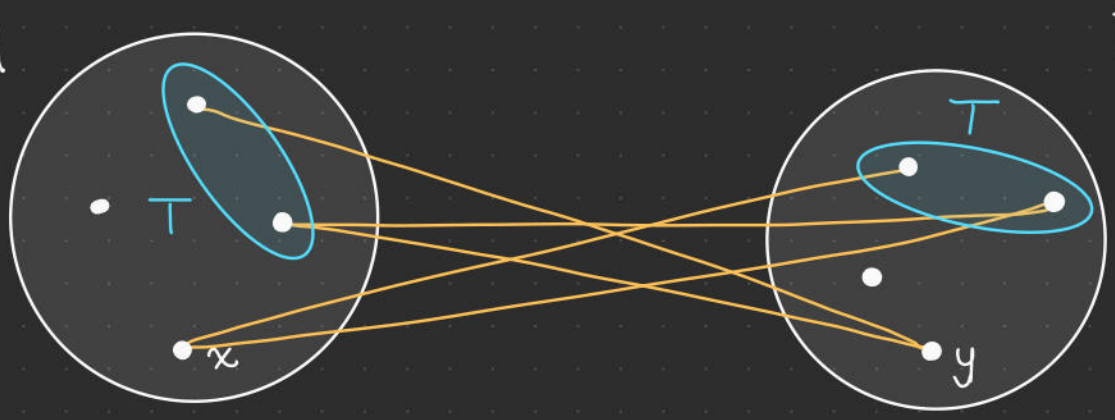
Break vertex sets up further to make counting easier.

If every vertex in U is adjacent to every vertex in $\bar{U}$ (complete), then

$$|(U, \bar{U})| = |U| \cdot |\bar{U}| \geq \overbrace{n-1}^{\text{from } \kappa(G) \leq n-1} \geq \kappa(G).$$

(want a bound to lazy count)

Otherwise, we choose an $x \in U$ and $y \in \bar{U}$, with x not adjacent to y.



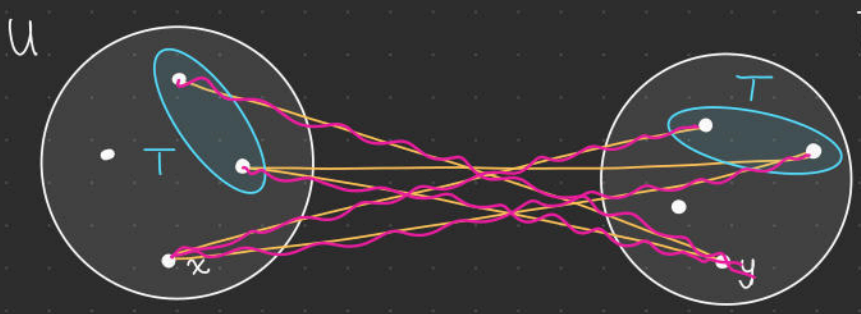
(only requirement is that x not adjacent to y, other things such as above are allowed)

Let T be the set of all vertices adjacent to x in $\bar{U}$, and all vertices in $U \setminus \{x\}$ adjacent to vertices in $\bar{U}$.

> We know we cant just get from x to y, but can take paths indirectly. Is there any way to get from x to y without going through T? No, there is not.

Every x-y path passes through T, so T is an x-y separating set.

Hence $|T| \geq \kappa(G)$ by definition.
If we choose the edges from x to $T \cap \bar{U}$, and one edge from $T \cap U$ to $\bar{U}$, we have



$|T|$ of edges of $(U, \bar{U})$.

Thus $k(G) \leq |T| \leq |(U, \bar{U})| = \lambda(G)$,
and so $k(G) \leq \lambda(G)$.